International Journal of Disaster Risk Reduction 108 (2024) 104528

Contents lists available at ScienceDirect

International Journal of Disaster Risk Reduction

journal homepage: www.elsevier.com/locate/ijdrr

Does willingness to pay for the traditional flood control measures *kasumitei* vary by river section of residential area?

Reiya Watanabe [a, b, *], Satoshi Kofuku [b], Kentaro Taki [c], Takehito Yoshida [d, e]

[a] Graduate School of Regional Resource Management, University of Hyogo, Hyogo, Japan
[b] Regional Co-creation Promotion Department, Institute of Environmental Informatics, IDEA Consultants, Inc., Kanagawa, Japan
[c] School of Environmental Science, University of Shiga Prefecture, Shiga, Japan
[d] Graduate School of Agricultural and Life Sciences, University of Tokyo, Tokyo, Japan
[e] Research Institute for Humanity and Nature, Kyoto, Japan

ARTICLE INFO

Keywords:
Ecosystem-based disaster risk reduction
Flood
Green infrastructure
Other effective conservation area-based measures
Paddy field

ABSTRACT

Kasumitei (open levees) comprise a traditional flood mitigation measure used in Japan. *Kasumitei* allow floodwater to flow into surrounding paddy fields, which act as temporary flood mitigation basins. After flooding, debris accumulated in paddy fields often brings crop damage and debris removal costs. Therefore, in some areas, the disaster mitigation functions of *kasumitei* may bring about conflicts between beneficiaries (downstream residents) and burden bearers (midstream residents). Ensuring the continued use of *kasumitei* in the future requires a mechanism to support the burden bearers. This study aimed to test the hypothesis that the willingness to pay (WTP) for maintaining the *kasumitei* would be lower for midstream residents (potential burden bearers) than for upstream (irrelevant) and downstream residents (potential beneficiaries). The results showed that contrary to the hypothesis, WTP for maintaining *kasumitei* was not affected by the river sections of the respondents' residence. Thus, a financial support scheme for maintaining *kasumitei* might be acceptable regardless of the beneficiary or bearer in the surveyed respondents. In addition, respondents with higher interest in ecosystem conservation tended to have higher WTP. Therefore, to gain support for the maintenance and use of the *kasumitei* (a type of ecosystem-based disaster risk reduction), the authorities would promote awareness of the measure's contribution to ecosystem conservation. Based on the results, we discussed the potential financial support schemes for maintaining the *kasumitei*.

1. Introduction

Since the 1950s, the frequency and intensity of heavy rainfall have been increasing worldwide, mainly owing to anthropogenic climate change and resulting in frequent human and economic losses [1,2]. Japan, located in the Asian monsoon region, is vulnerable to flood damage, especially in areas with steep topography and small river basins that receive high precipitation [3]. For example, Typhoon Hagibis in October 2019 recorded the highest precipitation in the Kanto, Koshin, and Tohoku regions, resulting in disasters caused by river overflows and mudslides [4]. Indeed, the largest ever heavy rainfall has been recorded every year, resulting in large-scale disasters, for which countermeasures are urgently needed [5]. However, efforts to maintain or build new gray infrastructures, such as levees and dams, are impeded by declining tax revenues resulting from the country's declining birthrate and aging population

* Corresponding author. Graduate School of Regional Resource Management, University of Hyogo, 128, Shounji, Toyooka, Hyogo, 668–0814, Japan.
E-mail address: watanabe.reiya.sw@alumni.tsukuba.ac.jp (R. Watanabe).

https://doi.org/10.1016/j.ijdrr.2024.104528
Received 26 October 2023; Received in revised form 18 March 2024; Accepted 30 April 2024
Available online 3 May 2024
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[5]. Under these domestic trends, interest is growing in ecosystem-based disaster risk reduction (Eco-DRR), a method that seeks to build a sustainable society that is resilient to natural disasters by avoiding the use of natural disaster-prone land to mitigate exposure to disasters and by utilizing the multiple functions of ecosystems [6,7]. Typical examples of Eco-DRR are the prevention of surface collapse and sediment runoff by conserving mountain forests, flood mitigation by maintaining wetlands, and tsunami and storm surge attenuation by safeguarding coastal forests [8]. The Fifth Basic Environmental Plan, which was approved by the Cabinet in April 2018, established the national policy to promote Eco-DRR [9]. The National Spatial Strategy (National Plan), which was approval by the Cabinet in 2023, also sets the goal of "creating a unique national land that nurtures a world-class beautiful nature and diverse culture" and including Eco-DRR as a measure to achieve this goal [10].

Kasumitei are discontinuous and open levees that leave an opening in the levees and extend the levees downstream of the opening toward the inner levee land, thereby doubling as the levee upstream of the opening [11]. Its exact origin is unknown, but according to an Edo period book, it was invented by a warrior, Shingen Takeda (1521–1573), during the Sengoku period [12]. The function of *kasumitei* varies depending on its location, but these levees are known to reduce river water levels by directing river water from the opening to the surrounding paddy fields, promote the elimination of inland water, return flood waters to the main stream, and alleviate levee piping [13]. Because of these features, *kasumitei* are positioned as a type of Eco-DRR, despite being a traditional flood mitigation measure [14,15]. *Kasumitei* are distributed in rivers throughout Japan from Hokkaido to the Kyushu region, with higher densities in rivers in the Hokuriku and Chubu regions [12]. The number of *kasumitei* in the Japanese 48 river systems was reported to be 6.2 ± 6.0 [1–24] (mean ± SD [min-max]) [16]. Taki et al. (2018) classified the disaster mitigation functions of *kasumitei* in the Kitagawa River basin into six categories: river level reduction, piping mitigation, storage, elimination of inundation flow and inland water, flood inundation area limitation, and inundation flow velocity control [17]. They compared the results of runoff and inundation calculations between the cases where *kasumitei* are preserved and where they are converted to a continuous levee. Taki et al. (2018) also quantitatively evaluated their disaster mitigation functions. The disaster mitigation functions of *kasumitei* have also been evaluated in the Kurobe River basin [18], Kinugawa River basin [19], Matsuura River basin [20], and Johara River basin [21]. In addition, the agricultural ditches and small rivers connected to the *kasumitei* are secured in continuity with the mainstream and serve as a temporary refuge for fish during floods and droughts [13,22]. In the Kitagawa River basin, fish diversity is higher in agricultural ditches and small streams connected to *kasumitei* compared with those that are disconnected [23]. Therefore, *kasumitei* is important for the flood mitigation and biodiversity conservation in floodplains.

Methods to use farmlands and floodplains as retarding basins, similar to *kasumitei*, exist in other countries as well [24–26]. For example, the discontinuous levee is used as to protect some inhabited land or farming areas against floods in Po River basin, Italy [27]. After the severe flood damage in 1995, the monkey cheek project has been conducted to lower the peak flood and reduce the flood risk by connecting a retarding basin to the river or waterway and discharging the rainwater after the water level in the river or waterway drops in Thailand's Chao Phraya River [28]. The preservation or restoration of floodplains are considered by the EU Floods Directive as an effective tool for reducing flood risks [29]. In Danube River, the restoration of floodplains had significant positive effects on reducing downstream flood risks [30]. Meanwhile, when residences are located near floodplains, the residents can be burden bearers of flood mitigation, whereas the downstream residents who receive its mitigation function can be beneficiaries [31,32]. Therefore, to maintain floodplains as retarding basins, a mechanism to support the burden bearers is necessary. In Austria, when farmlands are used as retarding basin for flood mitigation, the government compensates for land owners by three options: 1. establishment of an easement, 2. purchase of the inundated land, or 3. exchange for another land [33]. As the compensation for an easement, costs for crop damage, clean up, and disposal of floating debris can be paid out once in advance or in the event of a flood incident [33].

For the *kasumitei*, after a river overflows, driftwood and other debris accumulates in paddy fields surrounding the *kasumitei*, causing crop damage and requiring debris removal. The frequency and extent of flooding are not constant, and the range of benefits from the disaster mitigation function of the *kasumitei* has not been fully considered. The government does not have a compensation system for crop damage and debris removal, and farmers use a voluntary agricultural insurance system [34]. The *kasumitei* are often constructed in midstream such as those in the Kitagawa River basin, Fukui Prefecture [17]. Therefore, in some areas, a conflict may arise between beneficiaries (mainly downstream residents) and burden bearers (mainly midstream residents) of the disaster mitigation functions provided by the *kasumitei* [35]. To ensure the continued use of the *kasumitei* in the future, the government should formulate a mechanism to support the burden bearers. However, no previous study has quantitatively evaluated the value gap between the beneficiaries and burden bearers of *kasumitei* and presented a financial support scheme for maintaining *kasumitei* (except for a brief report [36] and an economic valuation of paddy fields near *kasumitei* [37]). In the present study, the disaster mitigation function of *kasumitei* was considered in relation to the river section where the residence is located as follows: upstream: areas where flood risk remains the same between with and without *kasumitei*; midstream: areas that would be flooded if *kasumitei* are present (potential burden bearers); downstream: areas where flood risk is reduced if *kasumitei* are present (potential beneficiaries). To consider the financial support scheme for maintaining *kasumitei*, we conducted a questionnaire survey among residents of three river sections (upstream, midstream, and downstream) and determined whether the willingness to pay (WTP) for maintaining the *kasumitei* vary among river sections, especially between downstream and midstream residents. We tested the hypothesis that the WTP would be lower for midstream residents (potential burden bearers) than for upstream (irrelevant) and downstream residents (potential beneficiaries).

2. Materials and methods

2.1. Design of online questionnaire

The online survey was conducted from January 27 to March 3, 2021, using the Rakuten Insight, Inc. online questionnaire system. The online questionnaire consisted of two parts. In the first part, we explained the disaster mitigation and other environmental func-

tions of the *kasumitei* and potential conflict between beneficiaries and burden bearers, using the Kitagawa River basin in Fukui Prefecture, Japan as an example. The second part included the WTP questions.

2.2. The first part: prior explanations for WTP questions

At first, we explained the fundamental information about the Kitagawa River as follows. This river flows through Takashima City, Shiga Prefecture, and Wakasa Town and Obama City, Fukui Prefecture (Fig. 1). The total length of the mainstream is 30.3 km, and its watershed area is 210.2 km^2. The river is relatively short and has a steep gradient, and the water rises rapidly after heavy rainfall especially in winter, summer, and typhoon seasons. Indeed, the mainstream and its tributaries were recorded as being flooded by typhoons from the 1950s–2000s, causing damage such as destruction and runoff of buildings [38]. For example, a typhoon in 1953 killed 53 people and caused extensive damage to approximately 1950 buildings.

Then, we explained the fundamental information about the *kasumitei* and its disaster mitigation and other environmental functions as follows. During floods, the *kasumitei* reduce river water levels by directing river water from the opening to the surrounding paddy fields, promote the elimination of inland water, return flood waters to the mainstream and the water that has overflowed upstream back to the downstream river, and backflow through the openings to inundate the surrounding areas, thereby reducing the risk of downstream flood and levee collapse. In the paddy fields surrounding the *kasumitei*, the diversity and abundance of fish and other organisms are high, which contributes to improve the habitat quality for endangered animals such as the Oriental white stork, *Ciconia boyciana* Swinhoe, 1873. If a river is divided into three sections: upstream, midstream, and downstream, the *kasumitei* are generally constructed in the midstream. In the Kitagawa River basin, 11 *kasumitei* are located in the midstream (9 in the Kita River, and 2 in the tributaries) [39].

After above explanations, we asked which of river sections applies to respondent's residence in relation to a neighboring river: 1. upstream, 2. midstream where *kasumitei* can be installed, 3. downstream which might be inundated by flooding, or 4. Not living near a river. In our study, the river section was used as an indicator whether they are a beneficiary (downstream), a burden bearer (midstream), or irrelevant (upstream) to the *kasumitei*. Residents of Fukui Prefecture where the Kitagawa River is located were the main respondents, but in order to have an enough number of respondents for the three river sections, residents of the neighboring prefectures of Ishikawa and Kyoto were also included (Fukui: n = 1228; Ishikawa: n = 44; Kyoto: n = 228). The total sample size was 1500 (including samples of preliminary tests: see Section 2.3) with 500 samples for each river section (upstream, midstream, and downstream).

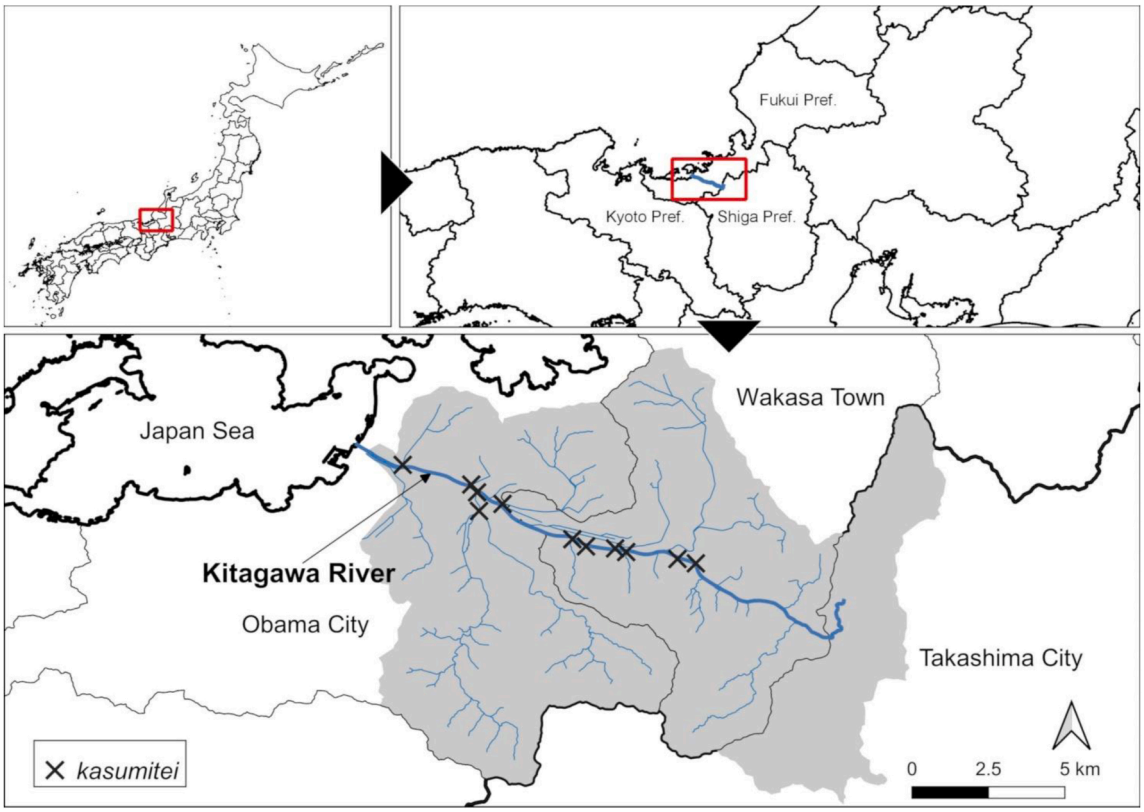

Fig. 1. Map of the Kitagawa River used for the prior explanation of WTP questions. The gray-filled area is the watershed boundary of the Kitagawa River basin.

2.3. The second part: WTP questions

To quantify the economic value of *kasumitei*, we used a double-bounded dichotomous choice (DBDC) CVM [40] (Fig. S1). The first question presented one of the three scenarios (Scenarios 1, 2, and 3) and a control scenario (Scenario 0), and explained the advantages and disadvantages of each scenario in tables and a figure (Tables 1 and 2; Fig. 2). The respondents were asked to agree, disagree, or neither with the scenario (Scenarios 1, 2, and 3). If they answered "disagree," they were asked to choose from the following three options: 1. because I do not see the need for improvement, 2. because I disagree with the tax structure, or 3. other reasons. The questionnaire then presented the annual taxable bid (JPY/family/year) for the scenario (Scenarios 1, 2, and 3) and asked if they were willing to pay it. The first bid values were set to b1 = {500, 1,000, 3000}. In the second question, respondents who indicated a willingness to pay in the first question were offered a higher bid than b1, and those who were not willing to pay were offered a lower bid than b1. The second bid values were set to b2 (b1: yes/no) = {500: 1000/100, 1000: 3000/500, 3000: 5000/1000}. To assess the validity of the bid value against range bias [41], we conducted a preliminary survey for respondents in Fukui Prefecture (n = 100 for each first bid and the total sample size was 300). We then confirmed that the confidential intervals of the median and truncated WTP values of the full model ranged within the offered bid value for each scenario (Fig. S2), as recommended by Kuriyama et al. (2013) [42]. Thus, we did not change the bid values and questionnaire design, and preliminary survey samples were included in the final statistical analysis. Before undertaking the statistical analysis, we omitted samples that disagreed with the scenario and chose option 2 (because I disagree with the tax structure) as the reason, or that disagreed with the scenario and chose option 1 (because I do not see the need for improvement) as the reason but agreed to pay for b1, because they were protest responses that chose "no" for other non-economic reasons, such as not being satisfied with the situation or the means of payment according to Jorgensen et al. (1999) [43]. Ultimately, scenarios 1, 2, and 3 had the following respective samples: 1378, 1424, and 1427.

We examined the potential factors affecting WTP according to previous studies (Table 3): the river section (upstream, midstream, or downstream), residential area (urban, rural, or other), annual income, sex, age, and prefecture (Fukui, Ishikawa, or Kyoto). The survey also included items related to the level of interest in flooding, green infrastructure, and ecosystem conservation (Tables 3 and 4). The level of interest was calculated by dividing the sum of the results of the responses to each related question by the number of questions, with higher values indicating a higher level of interest. To determine the level of "interest in ecosystem conservation" of respondents, we explained the conservation efforts for *C. boyciana* as follows. The Kunitomi district of Obama City, located downstream of the Kitagawa River, was one of the last breeding sites of *C. boyciana* before it became extinct in the wild in Japan. As such, the area has a high awareness of conservation activities, and efforts are conducted by the local government and local residents to install artifi-

Table 1
Description of four scenarios.

Scenario	Explanation	Breakdown of tax increase
0 (control)	Closing *kasumitei* and altering into a continuous levee	None
1	Closing *kasumitei* and altering into a larger and stronger continuous levee than that in scenario 0	Cost of renovation to continuous levees and maintenance of *kasumitei*
2	Maintain *kasumitei* in their current state	Cost of maintaining *kasumitei* and removing debris that flows into surrounding paddy fields
3	Maintaining *kasumitei* and conducting environmentally friendly farming on paddy fields in the hinterland of *kasumitei*	Cost of maintaining *kasumitei* and removing debris that flows into surrounding paddy fields Cost of human resource development and improvement of agricultural technology to promote environmentally friendly farming

Table 2
Advantages and disadvantages of each scenario.

Stakeholders	Scenario 0 (control)	Scenario 1	Scenario 2	Scenario 3
Overall (upstream, midstream, downstream)	× The feeding site of *C. boyciana* will disappear	× The feeding site of *C. boyciana* will disappear	- The feeding site of *C. boyciana* remains unchanged	○ The feeding site of *C. boyciana* will be increased by 20 %
Midstream: around the *kasumitei*	○ Withstands heavy rainfall approximately once every 30 years × Increases the likelihood of dike failure by 20 % × The flooded area will increase by 30 % × 1.5 times longer time required to drain rainwater	○ Withstands heavy rainfall approximately once every 50 years × Increases the likelihood of dike failure by 20 % × The flooded area will increase by 50 % × 1.5 times longer time required to drain rainwater	- Withstands heavy rainfall approximately once every 15 years - The possibility of levee failure does not change - The flooded area does not change - The time required to drain rainwater does not change	- Withstands heavy rainfall approximately once every 15 years - The possibility of levee failure does not change - The flooded area does not change - The time required to drain rainwater does not change
Downstream	○ Withstands heavy rainfall approximately once every 30 years × The flooded area will increase by 30 %	○ Withstands heavy rainfall approximately once every 50 years × The flooded area will increase by 50 %	- Withstands heavy rainfall approximately once every 15 years - The time required to drain rainwater does not change	- Withstands heavy rainfall approximately once every 15 years - The time required to drain rainwater does not change

○: advantage, × : disadvantage, -: neither.

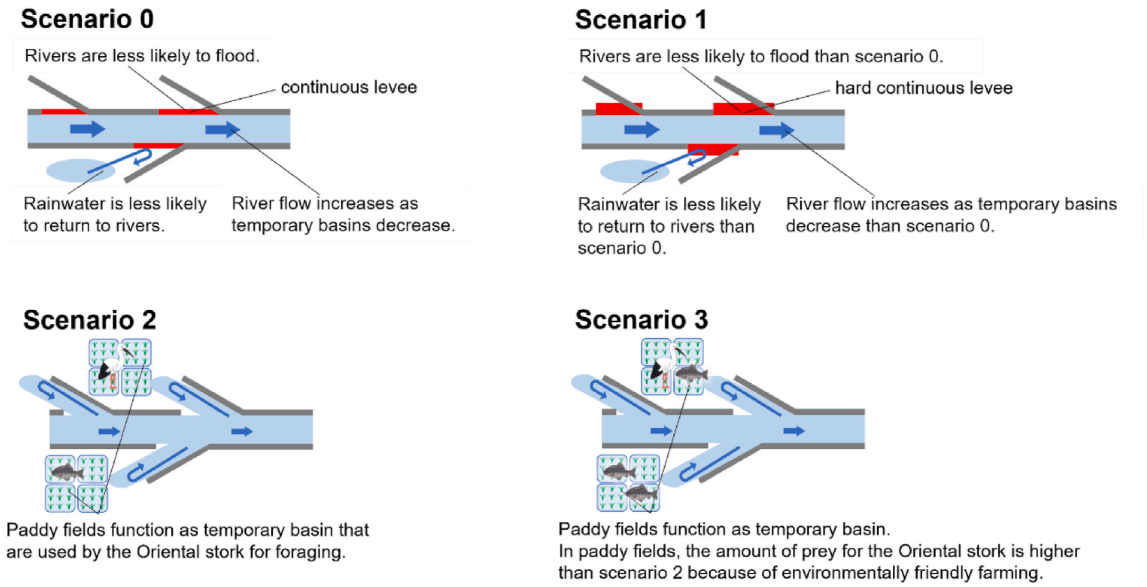

Fig. 2. Graphical abstract of each scenario.

cial nesting towers and create feeding habitats [44]. In the Kitagawa River basin, the paddy fields surrounding the *kasumitei* are used as feeding habitats for *C. boyciana* [45]. Because *C. boyciana* is a flagship species for ecosystem conservation in the Kitagawa River basin, our questions on the level of interest in ecosystem conservation were focused on the conservation of this species.

2.4. Statistical analysis

We used the statistical software R version 4.1.0 for all analyses [58]. To estimate WTP for the three scenarios, we fitted a DBDC CVM with log-logistic distribution using the dbchoice function of the Dcchoice package [59]. In the model, the DBDC response (yy: yes/yes, yn: yes/no, ny: no/yes, nn: no/no) was included as the response variable, whereas the river section, residential area, annual income, sex, age, prefecture, and level of interest (in flooding, green infrastructure, and ecosystem conservation) were included as explanatory variables. The natural logarithm of the total bids (log(b1+b2)) was included as a bid variable. To check for multicollinearity among the explanatory variables, we calculated the variance inflation factor (VIF) by using the check_ collinearity function in the 'performance' package [60]. No significant multicollinearity was found; VIF was less than five for all variables [61]. Thus, the model with all explanatory variables was used as the full model. We created models for all possible combinations of the explanatory variables, and calculated the Akaike information criterion (AIC) value for each model. The model with the lowest AIC value was selected as the best model.

We estimated the median and truncated mean WTPs using a maximum likelihood estimator for the three scenarios. To determine whether the median and truncated mean WTPs of the best model differed among the three scenarios, we calculated the 95 % confidence intervals of the WTPs using the bootstrap method (1000 replicates) and assessed whether the confidence intervals overlapped among the scenarios.

3. Results

3.1. Respondents of the questionnaire survey

Of the total respondents (n = 1500), 59 % were male, and more than 60 % were 30–50 years old (Table S1). The fraction of 30–50 years old in the population of Fukui, Ishikawa, and Kyoto Prefectures was approximately 30 % (Table S1) [62–64]. The sample bias toward 30 to 50-years-old adults might be because of their interests in registering as online survey monitors (Table S1). The residential area was almost equal to three categories: 27.5 % in urban (n = 412), 34.7 % in rural (n = 520), and 37.9 % in other (n = 568).

3.2. Percentage of agreement with scenario

Across all scenarios, the percentage of yy (yes/yes) responses (pay both b1 and b2) decreased as the value of b1 increased (Table S2), and the percentage of nn (no/no) responses (not pay both b1 and b2) increased as the value of b1 increased (Table S2), indicating that respondents agreed or disagreed to the payment scheme by considering the value of the bid offered. Also, the percentage of yy responses was higher in the order of scenario 3, 2, and 1, and that of nn was lower in the order of scenario 3, 2, and 1 across all first bid value.

Table 3
Description of explanatory variables.

Variable (type)	Definition	Mean ± SD [Min–Max]	Hypothesized effects on WTP based on previous studies
River section (CA)	River section of residence (1 = upstream, 2 = midstream, 3 = downstream, 4 = neither)	2.0 ± 0.8 [1–3]	The WTP would be lower in midstream residents (potential burden bearers) than upstream (irrelevant) and downstream residents (potential beneficiaries) because previous studies have demonstrated the perceptions on flood mitigation differ with location of residence relation to a neighboring river [46,47].
Residential area (CA)	Residential area of residence (1 = urban: area with developing commerce and distribution and high population density, 2 = rural: area with developing agriculture, forestry, and fishery and low population density, or 3 = other area: area other than urban and rural)	2.1 ± 0.8 [1–3]	The WTP would be lower in rural residents than residents in urban and other areas because previous studies have demonstrated the perceptions on flood mitigation differ with the land use patterns of residence [35,47,48].
Sex (CA)	Male (1) or female (2)	1.4 ± 0.5 [1–2]	The WTP would differ with sex because previous studies have demonstrated both positive and negative effects of male (or female) on flood risk reduction [49–53].
Annual income (CA)	Participant's household income. From less than 2 million JPY to over 18.01 million JPY; 10 categories in 2 million JPY increments (Range 1–10)	3.6 ± 1.8 [1–10]	The WTP would increase with annual income because the respondents would take their income into account [47,54,55].
Age (CO)	Years of age	47.3 ± 13.6 [17–85]	The WTP would increase with age because previous studies have demonstrated that older people has a high perception on flood risk [50,56].
Prefecture (CA)	Prefecture of residence (1 = Fukui, 2 = Ishikawa, 3 = Kyoto)	1.3 ± 0.7 [1–3]	The WTP would differ with prefecture of residence because respondents in those prefectures would have different environmental and socio-economic situations.
Interest in flooding (D)	"sum of responses to Q1 and Q2 in Table 3"/2. Higher values indicating a higher level of interest.	3.4 ± 0.9 [1–5]	The WTP would increase with interest in flooding because previous studies have demonstrated positive relationships between the perception on flood risk and WTP for flood mitigation [46,49,57].
Interest in green infrastructure (D)	"sum of responses to Q3 and Q4 in Table 3"/2. Higher values indicating a higher level of interest.	1.2 ± 0.5 [1–3]	The WTP would increase with interest in green infrastructure because previous studies have demonstrated positive relationships between the perception of effectiveness of green infrastructure against flood and the desire for green infrastructure [49].
Interest in ecosystem conservation (D)	"sum of responses to Q5–10 in Table 3"/6. Higher values indicating a higher level of interest.	2.9 ± 0.6 [1–4.5]	The WTP would increase with interest in ecosystem conservation because previous studies have demonstrated positive relationships between ecosystem conservation and the perception on flood risk [47,49].

Abbreviations: categorical variable (CA), continuous variable (CO), discrete variable (D).

3.3. WTP

The median WTP was significantly higher in scenarios 2 (lower–upper confidence bound: 824–1070) and 3 (1036–1324) compared with scenario 1 (570–784), and the truncated mean WTP was significantly higher for scenario 3 (191–2158) compared with scenario 1 (1596–1800) (Fig. 3).

3.4. Factors influencing WTP

Table 5 gives the coefficients of the best model for each scenario. For scenario 1, the best model had residential area, annual income, prefecture, and level of interest in ecosystem conservation as explanatory variables. Respondents with a high annual income ($p < 0.01$) and high level of interest in ecosystem conservation ($p < 0.01$) were more likely to agree with scenario 1. Respondents in Kyoto Prefecture were more likely to agree with scenario 1 compared with those in Fukui Prefecture ($p = 0.03$).

For scenario 2, the best model had annual income, age, sex, prefecture, and level of interest in ecosystem conservation as explanatory variables. As also shown in Table 5, respondents with a high annual income ($p = 0.02$), age ($p < 0.01$), and level of interest in ecosystem conservation ($p < 0.01$) were more likely to agree with scenario 2. Respondents in Kyoto Prefecture were more likely to agree with scenario 2 compared with those in Fukui Prefecture ($p < 0.01$).

For scenario 3, the best model had residential area, annual income, age, prefecture, and level of interest in ecosystem conservation as explanatory variables. Respondents with a high annual income ($p < 0.01$), age ($p < 0.01$), and level of interest in ecosystem conservation ($p < 0.01$) were more likely to agree with scenario 3. Respondents in Kyoto Prefecture were more likely to agree with scenario 3 compared with those in Fukui Prefecture ($p < 0.01$).

For all scenarios, the percentage of respondents in favor of the scenarios decreased as the natural logarithm of the amount offered increased ($p < 0.01$), as shown in Table 5. Contrary to the hypothesis, for all scenarios, the river section was not an explanatory variable of the best model, and the effect of the river section was not significant in the full model (Tables 5 and 6).

Table 4
Questions on interest in flooding, green infrastructure, and ecological conservation.

Interest	Question	Options
Flooding	Q1. Are you interested in flood disaster (flooding, inundation, etc.)? Please select an option.	**1** Not interested at all **2** Not very interested **3** Can't say either way **4** Somewhat interested **5** Very much interested
	Q2. How do you feel about the risk of flooding in your community (town or district unit)? Please select an option.	**1** Not at all threatened **2** Not very concerned **3** Can't say either way **4** Somewhat threatened **5** Very threatened
Green infrastructure	Q3. Do you know about *kasumitei*? Please select an option.	**1** I did not know. **2** I had heard of it. **3** I knew about it.
	Q4. Infrastructure that takes advantage of the diverse functions and mechanisms of ecosystems and green spaces, such as *kasumitei*, is called "green infrastructure." Do you know what green infrastructure is? Please select an option.	**1** I did not know. **2** I had heard about it. **3** I knew about it.
Ecosystem conservation	Q5. Do you know about the efforts to reintroduce *C. boyciana* in Fukui Prefecture? Please select an option.	**1** I did not know. **2** I had heard about it **3** I knew about it
	Q6. Are you interested in the reintroduction of *C. boyciana* to Fukui Prefecture? Please select an option.	**1** Not interested at all **2** Not very interested **3** Can't say either way **4** Somewhat interested **5** Very much interested
	Q7. How do you feel about conserving the natural environment? Please select an option.	**1** Not interested at all **2** Not very interested **3** Can't say either way **4** Somewhat interested **5** Very much interested
	Q8. Are natural places important to you? Please select an option.	**1** Not at all **2** Not so much **3** Can't say either way **4** Somewhat agree **5** Very much so
	Q9. In the past two years, have you participated in any nature conservation activities (e.g., river clean-ups, plant and animal observations, conservation activities)? Please select an option.	**1** Not at all **2** Once or twice **3** Three or four times **4** More than five times

(*continued on next page*)

Table 4 (*continued*)

Interest	Question	Options
	Q10. Would you like to volunteer in the conservation and management of *C. boyciana* habitat? Please select an option.	**1** Not at all **2** Not so much **3** Can't say either way **4** Somewhat agree **5** Very much so

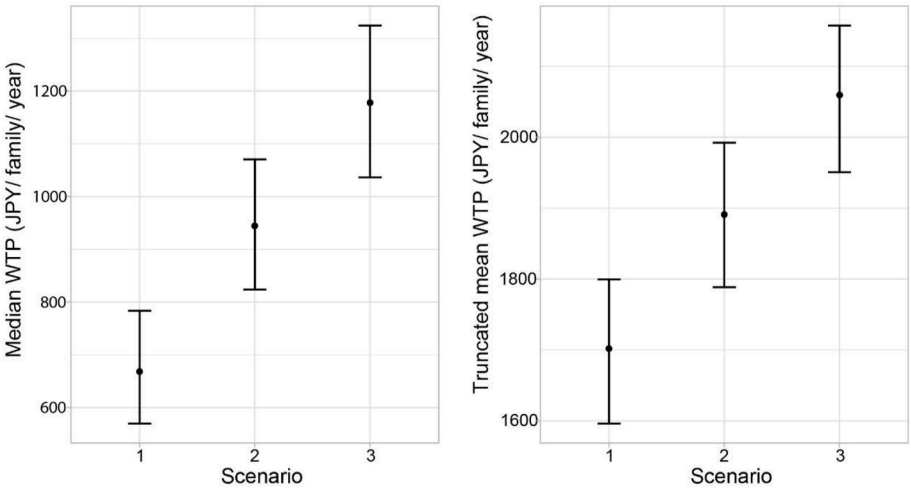

Fig. 3. Comparison of median and truncated mean willingness to pay (WTP) of the best model among scenarios. Error bars indicate 95 % confidence intervals by 1000 bootstraps. JPY: Japanese yen.

4. Discussion

4.1. Factors influencing the value of kasumitei

We hypothesized that WTP for maintaining *kasumitei* would be lower for midstream resident (potential burden bearers) than for upstream and downstream residents (potential beneficiaries). To the best of our knowledge, this is the first study to evaluate whether the value of Eco-DRR would differ by residential area. However, contrary to the hypothesis, WTP for maintaining *kasumitei* was not affected by the river section of residence. In our study, the river section was used as an indicator whether they are a beneficiary (downstream), a burden bearer (midstream), or irrelevant (upstream) to the *kasumitei*. Thus, a financial support scheme for maintaining *kasumitei* might be acceptable regardless of the impacts (benefits or burden) to the residents in surveyed respondents. Although we sufficiently explained the relation between river section and benefit of the *kasumitei* during the questionnaire survey, the respondents may have been unaware of the beneficiaries and burden bearers of the *kasumitei*, especially among respondents from outside of the *kasumitei* neighborhood. Indeed, Roder et al. (2019) reported that respondents living close to major rivers are associated with a higher threat appraisal for flooding [46]. Another possible reason for the consistent WTP among the river sections is that only some farmers considered the *kasumitei* a burden, whereas those who felt that it was a beneficial system were a thin and broad segment of the population. Thus, in order to detect the difference of WTP between beneficiaries and burden bearers of the *kasumitei*, it would be necessary to incorporate the detailed location of residence and occupation (especially whether rice farmer or not) of respondents as explanatory variables in the WTP model.

In all scenarios, WTP was higher among those with high annual incomes and interest in ecosystem conservation. Our finding on the increase in WTP with annual income is similar to that reported for the perceptions of flood risk [47,54,55], and can be taken as evidence that respondents took their income into account when answering the questions. According to a study that assessed consumers' price premium for ecosystem and environment-friendly rice, consumers who were aware of environmentally friendly framings are more willing to pay, suggesting that policies that omit detailed explanations on the initiatives and focus on informing about the existence of the initiatives to consumers have some relevance [65]. Hori et al. (2023) conducted a questionnaire survey in Higashiomi City, Shiga Prefecture, and performed a path analysis of the factors influencing the intention to maintain and use *kasumitei* [49]. The results indicated that awareness of the existence of *kasumitei* strongly increased the perception of anti-flood effects and other benefits, and causally decreased the perception of disadvantages for the *kasumitei*. Our study also highlighted the importance of raising public awareness of not only the flood mitigation function but also the contribution to ecosystem conservation to gain support for the maintenance and use of *kasumitei*. In contrast, our respondents who were more concerned about ecosystem conservation were more likely to agree with scenario 1, in which *kasumitei* are upgraded into a strong continuous levee. Although the exact reason for this result is

Table 5
Coefficients of the best model with multinomial logit analysis for each scenario.

	Estimates	Standard error	Z value	*p value*
Scenario 1				
Intercept	**2.36**	0.32	7.40	<0.01
Residential area (Others)	−0.06	0.13	−0.49	0.6
Residential area (Rural)	0.18	0.13	1.38	0.2
Annual income	**0.09**	0.03	3.12	<0.01
Prefecture (Ishikawa)	**0.63**	0.30	2.12	0.03
Prefecture (Kyoto)	0.16	0.15	1.07	0.3
Interests in EC	**0.59**	0.09	6.88	<0.01
Log (b_1+b_2)	**−0.68**	0.03	−25.46	<0.01
Scenario 2				
Intercept	**1.57**	0.34	4.59	<0.01
Annual income	**0.06**	0.03	2.28	0.02
Age	**0.01**	0.004	2.73	<0.01
Prefecture (Ishikawa)	0.54	0.29	1.84	0.07
Prefecture (Kyoto)	**0.42**	0.14	2.97	<0.01
Interests in EC	**1.00**	0.09	11.38	<0.01
Log (b_1+b_2)	**−0.77**	0.03	−26.73	<0.01
Scenario 3				
Intercept	**1.83**	0.36	5.06	<0.01
Residential area (Others)	−0.15	0.13	−1.20	0.2
Residential area (Rural)	0.09	0.13	0.70	0.5
Annual income	**0.08**	0.03	3.02	<0.01
Age	**0.01**	0.004	2.64	<0.01
Prefecture (Ishikawa)	0.50	0.31	1.64	0.1
Prefecture (Kyoto)	**0.42**	0.14	2.94	<0.01
Interests in EC	**1.01**	0.09	11.36	<0.01
Log (b_1+b_2)	**−0.79**	0.03	−26.74	<0.01

Abbreviations: first bid (b1), second bid (b2), ecosystem conservation (EC). Significant results are shown in bold ($p < 0.05$). The estimates of residential area (city, rural, or others) are based on the category of "city."

unclear, we found that some of the respondents who were more concerned about ecosystem conservation were also more concerned about flooding, as interests in flooding and ecosystem conservation positively correlated ($r = 0.43$, $p < 0.01$; Table S3). In our study, interest in green infrastructure (including the prior knowledge on *kasumitei*) and that in flooding were not included in the best models across all scenarios. This result differs with previous studies that demonstrated positive relationships between the perception on flood risk and the WTP for flood mitigation, and those between the perception of effectiveness of green infrastructure against flood and the desire for green infrastructure [46,49,57]. Because we did not have the information of the detailed location of the respondent's residence, we could not include the factors related to actual flood risk (e.g., inundation depth and distance to a river) as explanatory variables in the models. These factors might affect the WTP for maintaining the *kasumitei*, and the further analysis are needed.

In scenario 3, WTP was higher for older people, which may indicate that older people had a better understanding of the flood mitigation function of *kasumitei*. This finding is consistent with previous studies that demonstrated that older people has a high perception on flood risk [50,56]. However, our samples had the bias toward 30 to 50-years-old, so that the samples of other age classes should be increased in order to confirm the trend with age. In scenarios 1 and 3, WTP was higher for residents of Kyoto Prefecture than those of Fukui Prefecture. In scenario 2, WTP was higher for residents of Ishikawa Prefecture than those of Fukui Prefecture. The survey respondents of our study who had registered in the Rakuten Insight, Inc. online survey system did not necessarily represent the all population of each prefecture. Although further research is needed to clarify the reasons for the differences in WTP by prefecture, our results may indicate that a financial support scheme need to be set in accordance with the situation in each prefecture.

WTP was highest in scenario 3 (maintaining the *kasumitei* and promoting environment-friendly farming in paddy fields in the hinterland), indicating that the respondents were generally cooperative in maintaining the *kasumitei*. In a study that interviewed farmers in the Matsukawa River basin, Saga Prefecture, flooding of the paddy fields in the hinterland of the *kasumitei* does not cause overturning of the rice plants nor extensive damage because the flooding lasts only a short time and leaves little mud on paddy fields; rather, such flooding has advantages because the soil becomes more fertile [35]. In other basins where *kasumitei* have been installed, the WTP should be calculated using the same method as in our study, and the maintenance of *kasumitei* should be considered by the responses of residents in each local community. Flood simulations with and without *kasumitei* have reported cases where peak downstream flows increase more when *kasumitei* are present than when they are not [34]. The main functions of the *kasumitei* in such cases are to promote the exclusion of inland water and shorten the flooding time, and the beneficiaries are the landowners and farmers in the hinterland of the *kasumitei*. Thus, the beneficiary structure of *kasumitei* is complex, and beneficiaries may even be difficult to identify in some areas. Our results, which showed a certain level of support for the maintenance of *kasumitei* regardless of the river section, were consistent with prior findings.

Table 6
Coefficients of the full model with multinomial logit analysis for each scenario.

	Estimates	Standard error	Z value	*p value*
Scenario 1				
Intercept	**2.02**	0.46	4.37	<0.01
River section (midstream)	−0.13	0.12	−1.05	0.29
River section (upstream)	−0.002	0.16	−0.01	0.99
Residential area (others)	−0.05	0.13	−0.40	0.69
Residential area (rural)	0.21	0.14	1.51	0.13
Annual income	**0.10**	0.03	3.37	<0.01
Sex	0.05	0.11	0.47	0.64
Age	0.01	0.004	1.28	0.20
Prefecture (Ishikawa)	0.61	0.32	1.88	0.06
Prefecture (Kyoto)	0.12	0.19	0.62	0.54
Interests in FD	0.08	0.06	1.30	0.19
Interests in GI	−0.08	0.12	−0.66	0.51
Interests in EC	**0.54**	0.10	5.51	<0.01
Log (b_1+b_2)	**−0.69**	0.03	−25.47	<0.01
Scenario 2				
Intercept	**1.47**	0.46	3.20	<0.01
River section (midstream)	−0.05	0.12	−0.39	0.70
River section (upstream)	0.03	0.16	0.18	0.86
Residential area (others)	−0.06	0.13	−0.50	0.62
Residential area (rural)	0.09	0.13	0.64	0.52
Annual income	**0.06**	0.03	2.17	0.03
Sex	0.05	0.11	0.42	0.68
Age	**0.01**	0.004	2.65	0.01
Prefecture (Ishikawa)	0.51	0.32	1.60	0.11
Prefecture (Kyoto)	**0.39**	0.19	2.05	0.04
Interests in FD	−0.02	0.07	−0.24	0.81
Interests in GI	0.10	0.12	0.91	0.36
Interests in EC	**0.98**	0.10	9.86	<0.01
Log (b_1+b_2)	**−0.77**	0.03	−26.73	<0.01
Scenario 3				
Intercept	**1.73**	0.46	3.74	<0.01
River section (midstream)	−0.06	0.12	−0.48	0.63
River section (upstream)	−0.15	0.16	−0.94	0.35
Residential area (Others)	−0.14	0.13	−1.13	0.26
Residential area (Rural)	0.12	0.13	0.91	0.37
Annual income	**0.08**	0.03	2.97	<0.01
Sex	0.06	0.11	0.55	0.58
Age	**0.01**	0.004	2.57	0.01
Prefecture (Ishikawa)	0.60	0.33	1.81	0.07
Prefecture (Kyoto)	**0.53**	0.19	2.76	0.01
Interests in FD	−0.02	0.06	−0.26	0.80
Interests in GI	0.08	0.12	0.68	0.50
Interests in EC	**1.01**	0.10	10.06	<0.01
Log (b_1+b_2)	**−0.79**	0.03	−26.74	<0.01

Abbreviations: first bid (b1), second bid (b2), Flooding (FD), Green Infrastructure (GI), Ecosystem Conservation (EC). Significant results are shown in bold ($p < 0.05$). The estimates of river section (downstream, midstream, or upstream) and residential area (city, rural, or others) are based on the categories of "downstream" and "city," respectively.

4.2. Potential financial support schemes for maintaining kasumitei

One of the financial instruments, payment for ecosystem services (PES) has commonly been used to finance biodiversity conservation and, increasingly, carbon storage and watershed or water resource management [66]. However, PES schemes are not yet widely used to capture Eco-DRR benefits [67]. In Japan, efforts similar to PES exists; conservation of upstream headwater forests by downstream residents has long been carried out in various regions [68]. For example, Yokohama City, Kanagawa Prefecture, has purchased about one-third of the forest in the upstream village of Doshi, Yamanashi Prefecture, and taxes collected from water bills are used to pay for the management of the forest [69]. It should be noted that PES can be applied when the beneficiaries and burden bearers of ecosystem services are clear [66]. We found that the WTP for maintaining the *kasumitei* did not change between potential burden bearers and beneficiaries. Thus, it would be difficult to simply apply PES for maintaining the *kasumitei* as the beneficiaries and burden bearers of the *kasumitei* could not be clearly identified.

Goods such as *kasumitei*, whose beneficiaries are difficult to identify and whose accessibility is difficult to restrict, are called public goods in the field of environmental economics [70]. A typical assumption is that their maintenance and management should be carried out by public agencies. Therefore, in the case of flooding, maintenance and management should be financed mainly by local taxes. Scholars have proposed several ways for bearing the costs of environmental policies, including the principle of the burden

borne by those who cause the burden on the environment, the burden borne by the beneficiaries of the policy, the burden borne by those who can pay, and participatory tax systems, such as a forest environmental tax [71]. In our study, the WTP was higher for respondents with high annual incomes, indicating the feasibility of applying a tax to maintain *kasumitei*.

Meanwhile, when the hinterland of *kasumitei* is flooded, debris often accumulates in the paddy fields, and debris removal requires a large amount of money. The variation in the frequency and extent of farmland flooding and the ambiguity of the range of benefits from the flood mitigation function of *kasumitei* have made it difficult for the government to budget for debris removal. In Austria, however, when farmlands are used as retarding basin for flood mitigation, the government compensate for land owners by three options: 1. establishment of an easement, 2. purchase of the inundated land, or 3. exchange for another land [33]. As the compensation for the easement, costs for crop damage, clean up, and disposal of floating debris can be paid out once in advance or in the event of a flood incident [33]. In contrast, the following challenges have been identified with this advanced Austrian government compensation measure. The compensation price is too low for landowners, and municipalities or associations often have to pay additional acceptance fees to obtain approval. One-time payment of damages to land caused by floods is preferred by most water authorities, although farmers perceived it as unfair because future development is uncertain and they would have to bear the risks alone.

We propose the potential financial support schemes for maintaining the *kasumitei*. The Ministry of the Environment has started to issue the certificates to "sites where biodiversity conservation is conducted through a broader range of stakeholders" as the Japanese "Other Effective Conservation area-based Measures" (OECM, *Shizen Kyosei Site* in Japanese) in 2023, involving the study of financial incentives for promoting its application [72]. The paddy fields in the hinterland of the *kasumitei* have been confirmed to be habitats of endangered floodplain fish species, such as the oily bitterling, *Tanakia limbata* (Temminck & Schlegel, 1846), and the Oriental weatherfish, *Misgurnus anguillicaudatus* (Cantor, 1842) [23]. Thus, the *kasumitei* and their hinterlands could satisfy the criteria for certifying sites as Japanese OECM. In according with the document presented by the Ministry of the Environment [73], local governments would apply the *kasumitei* and their hinterlands to a Japanese OECM and they would also mention it in the regional revitalization plan to receive the corporate version of the hometown tax payment (*Kigyouban Furusato Nozei* in Japanese), which can be used to maintain the *kasumitei* and compensate farmers for crop damage and debris removal costs. However, this initiative would require local governments with *kasumitei* to revise their local comprehensive strategies, such as regional revitalization plans.

5. Conclusion

We examined whether the WTP for maintaining the *kasumitei* would vary among river sections of residence, especially between downstream residents (potential beneficiaries) and the midstream residents (potential burden bearers). However, our results showed that WTP for maintaining *kasumitei* was not affected by the river section of residence. Thus, a financial support scheme for maintaining *kasumitei* might be acceptable regardless of the status of residents as beneficiary or burden bearer in surveyed respondents. Notably, the beneficiary structure of *kasumitei* is complex, and the beneficiaries are difficult to identify in some areas. Therefore, the financial support scheme needs to be set at the regional scale covering different river sections of residence. Our results showed that respondents with a higher level of interest in ecosystem conservation tended to have higher WTP. Therefore, to gain support for the maintenance and use of the *kasumitei*, it should be effective for the government to promote awareness of its contribution to ecosystem conservation. Finally, we recommend that local governments apply the *kasumitei* and their hinterlands to a Japanese OECM and they also mention it in their regional revitalization plans, to receive the corporate version of hometown tax payment, which can be used to maintain the *kasumitei* and to cover crop damage and debris removal costs.

Funding

This work was supported by the Research Institute for Humanity and Nature, Japan (RIHN: a constituent member of NIHU) project no. RIHN 14200103.

CRediT authorship contribution statement

Reiya Watanabe: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Satoshi Kofuku:** Conceptualization, Investigation, Methodology, Supervision, Writing – review & editing. **Kentaro Taki:** Validation, Writing – review & editing. **Takehito Yoshida:** Funding acquisition, Resources, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

We thank Dr. Ayumi Onuma (Professor at Keio University) and Dr. Takahiro Tsuge (Professor at Sophia University) for their constructive comments on the questionnaire design and statistical analysis methods involved in our study. We acknowledge Satoshi Ise (IDEA Consultants, Inc., Japan) for supporting to organize the data set and Editage (www.editage.com) for English editing.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijdrr.2024.104528.

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